Using Gossips to Spread Information

Theory and Evidence from Two Randomized Controlled Trials

Banerjee, Chandrasekhar, Duflo & Jackson (2019)

Replication Study | ECO613M

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Research Questions

Does seeding information through 'gossip nominees' lead to significantly greater information diffusion compared to random or status-based seeding?

How can we easily and cheaply identify highly central individuals without collecting network data?

Dataset

Two Randomized Controlled Trials (RCTs) conducted in rural India

- Karnataka RCT: Information diffusion via phone-based lottery
- Haryana RCT: Immunization awareness and uptake

Data Source: *https://academic.oup.com/restud/article/86/6/2453/5345571*

Theoritical Framework

Network gossip → diffusion centrality identification

The Model of Network Communication:

- Network w $\in [0,1]^{n\times n}$: w_{ij} = prob. i tells j,where w is the adjacency matrix. Unless stated o/w, w is taken as constant.
- Above stated relation needs not be reciprocal.

Diffusion centrality

- We define DC based on random information flow through network
- Piece of information originated at node i is heard by 'j' with probability w_{ij} and the process continues for T periods.
- DC measures extent of spread of information as function of initial node given by:

$$DC\left(w,T\right)=H\left(w,T\right).1=\left(\sum_{t=1}^{T}w^{t}\right).1$$

where, H(w,T) is the hearing matrix.

- Here, DC measures the expected number of times some informatrion generated at node i is heard by others.

Network Gossip

- DC is measured from sender's persepctive and NG is measured from receiver's persepective.
- We have defined heraing matrix as:

$$H\left(w,T\right)=\sum_{t=1}^{T}w^{t}$$

- Here, Network Gossip is defined by node j to be the column 'j' of the hearing matrix.
- $$NG(w,T)_{j}=H\left(w,T\right)_{.j.}$$

Thus NG_j lists the expected number of times a node j will hear a piece of news as a fucntion of the origin node of information.

Methodology

Regression Analysis:

$$y_j = \theta_0 + \theta_1 GossipTreatment_j + \theta_2 ElderTreatment_j + \theta_3 NumberSeeds_j + \theta_4 NumberGossip_j + \theta_5 NumberElder_j + u_j$$

- Estimates how assigning gossip or elder seeds(vs random) affects information diffusion

$$y_j = \theta + \beta_1 GossipReached_j + \beta_2 ElderReachedj + \beta_3 NumberSeedsj + \beta_4 NumberGossipj + \beta_5 NumberElderj + \varepsilon_j$$

- Measures the effect of actually reaching atleast one gossip or elder on diffusion

$$GossipReached_j = \pi_0 + \pi_1 GossipTreatment_j + \pi_2 ElderTreatment_j + \pi_3 NumberSeeds_j + \pi_4 NumberGossip_j + \pi_5 NumberElder_j + v_j$$

- Shows how treatment assignment predicts the likelihood of reaching a gossip(first stage)

$$ElderReached_j = \rho_0 + \rho_1 GossipTreatment_j + \rho_2 ElderTreatment_j + \rho_3 NumberSeeds_j + \rho_4 NumberGossip_j + \rho_5 NumberElder_j +$$

- Shows how treatment assignment predicts the likelihood of reaching an elder(first stage)

Study 1: Karnataka Cell Phone RCT

213 villages · 3 treatment arms · Information = free lottery call

Experimental Design

T1: Random
k = 3 or 5 random households seeded

T2: Gossip
Households nominated as 'good info spreaders'

T3: Elder
Recognized village authorities & leaders

Avg calls (Gossip villages)

8.1
Avg calls (Random villages)

6.9
Avg calls (Elder villages)

+22%
Gossip vs. Random (RF)

TABLE 1
Calls received by treatment

	(1) RF Calls received	(2) OLS Calls received	(3) IV 1: First Stage At least 1 gossip	(4) IV 2: First stage At least 1 elder	(5) IV: Second stage Calls received
Gossip treatment	3.651		0.644	0.328	
	(2.786)		(0.0660)	(0.0824)	
Elder treatment	−1.219		0.230	0.842	
	(2.053)		(0.0807)	(0.0509)	
At least 1 gossip		3.786			7.436
		(1.858)			(4.266)
At least 1 elder		0.792			−3.475
		(2.056)			(2.259)
Observations	212	212	212	212	212
Control group mean	8.077	5.846	0.391	0.184	5.805
Gossip treatment = Elder treatment (pval.)	0.0300		0	0	
At least 1 gossip = At least 1 elder (pval.)		0.330			0.0300

	(1) RF Calls received Seeds	(2) OLS Calls received Seeds	(3) IV 1: First stage At least 1 gossip	(4) IV 2: First stage At least 1 elder	(5) IV: Second stage Calls received
Gossip treatment	1.053		0.644	0.328	
	(0.698)		(0.0660)	(0.0824)	
Elder treatment	−0.116		0.230	0.842	
	(0.518)		(0.0807)	(0.0509)	
At least 1 gossip		0.952			1.979
		(0.501)			(1.071)
At least 1 elder		0.309			−0.677
		(0.511)			(0.588)
Observations	212	212	212	212	212
Control group mean	1.967	1.451	0.391	0.184	1.317
Gossip treatment = Elder treatment (pval.)	0.0400		0	0	
At least 1 gossip = At least 1 elder (pval.)		0.410			0.0400

Figure 1

CDF of calls

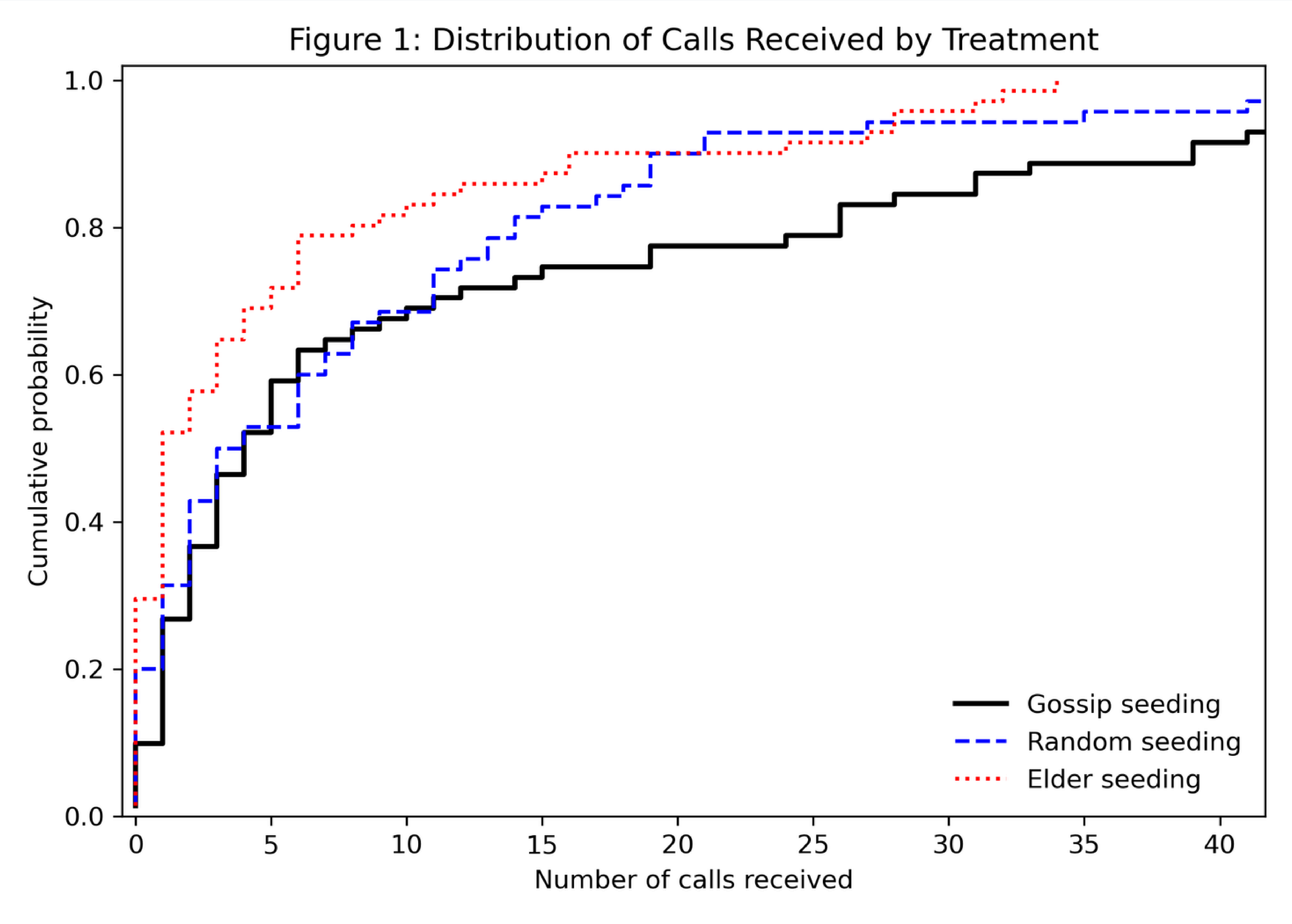


Table 1, Panel A

Calls received by treatments

```
=============== PANEL A: TOTAL CALLS ===============

Column 1: Reduced Form
              Coef.   Std.Err.
Intercept -1.162222  4.531769
gossip     3.593839  2.612923
elder     -2.009723  1.872812
num_seeds  1.118071  0.951762
gossip_ct -0.111916  0.178199
elder_ct   0.896347  0.377835

Control mean (random villages): 8.443

Column 2: OLS
                     Coef.   Std.Err.
Intercept        -1.925433  4.452546
atleast_gossip_1  4.067788  1.789964
atleast_elder_1   0.253896  1.876376
num_seeds         0.789884  0.940258
gossip_ct        -0.305339  0.191530
elder_ct          1.080416  0.398477

Column 3: First Stage (Gossip)
              Coef.   Std.Err.
Intercept -0.195607  0.115898
gossip     0.679132  0.057921
elder      0.233015  0.075071
num_seeds  0.068650  0.025823
gossip_ct  0.043503  0.005607
elder_ct  -0.007236  0.008028

Column 4: First Stage (Elder)
              Coef.   Std.Err.
Intercept  0.080916  0.110997
gossip     0.306066  0.076380
elder      0.834435  0.045693
num_seeds -0.034687  0.026057
gossip_ct  0.009250  0.006169
elder_ct   0.026628  0.009095
```

```
Column 5: IV
                              Parameter Estimates
==============================================================================
                  Parameter  Std. Err.     T-stat    P-value    Lower CI    Upper CI
------------------------------------------------------------------------------
const                0.6245     4.2254     0.1478     0.8825     -7.6570      8.9061
num_seeds            0.4630     0.9411     0.4920     0.6227     -1.3815      2.3076
gossip_ct           -0.3882     0.2229    -1.7415     0.0816     -0.8250      0.0487
elder_ct             1.0675     0.3940     2.7092     0.0067      0.2952      1.8398
atleast_gossip_1     7.2954     3.9032     1.8691     0.0616     -0.3548      14.946
atleast_elder_1     -4.4457     2.1765    -2.0426     0.0411     -8.7116     -0.1799
==============================================================================
```

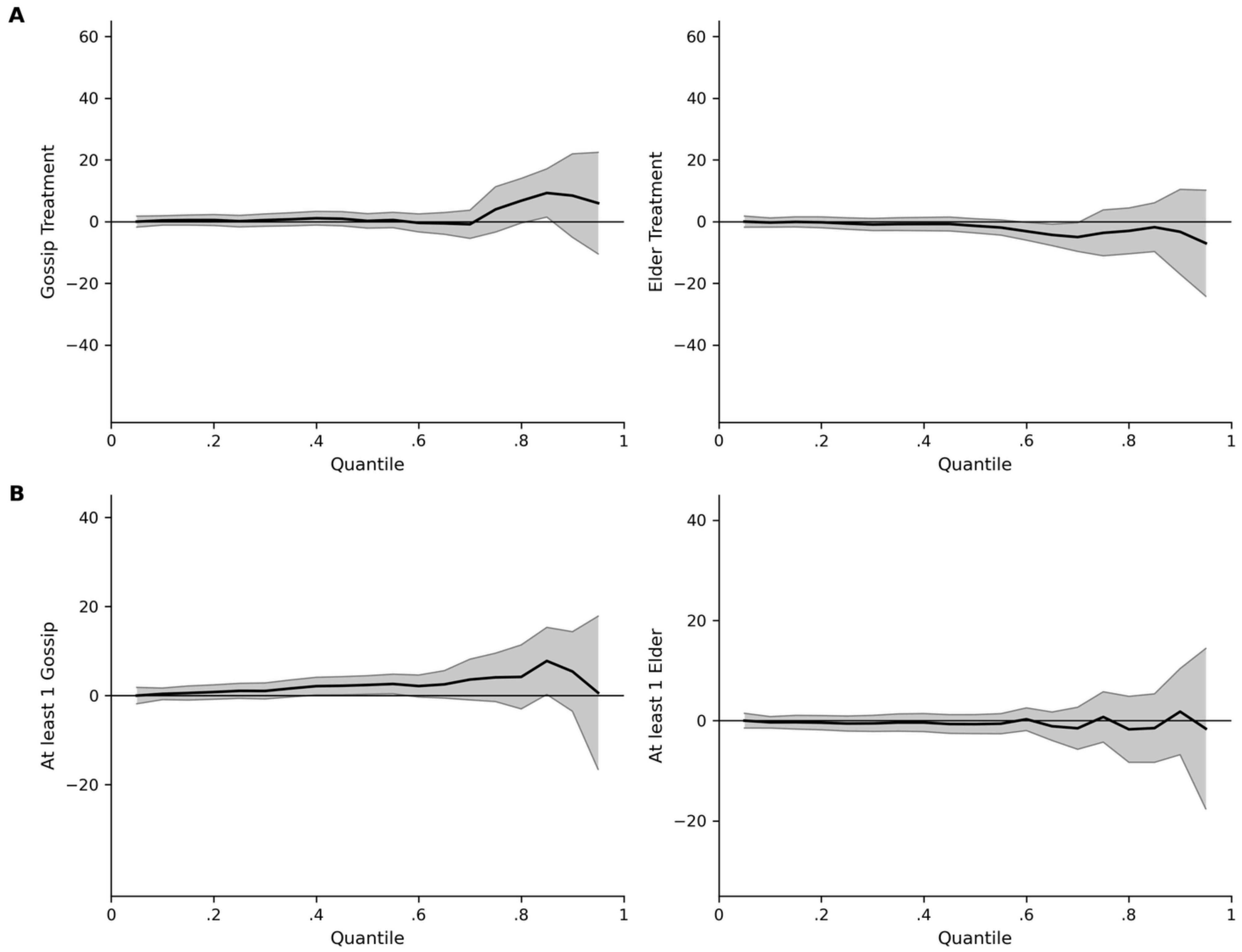
Table 1, Panel B

```
================ PANEL B: CALLS PER SEED ================

Column 1: Reduced Form
              Coef.   Std.Err.
Intercept  1.830542  1.131980
gossip     0.984906  0.659788
elder     -0.353610  0.486632
num_seeds -0.299728  0.251163
gossip_ct -0.026463  0.047498
elder_ct   0.247680  0.099460

Column 2: OLS
                     Coef.   Std.Err.
Intercept          1.689721  1.122936
atleast_gossip_1   0.997298  0.486918
atleast_elder_1    0.169523  0.475368
num_seeds         -0.376143  0.254130
gossip_ct         -0.077262  0.050026
elder_ct           0.290433  0.104358

Column 3: First Stage (Gossip)
              Coef.   Std.Err.
Intercept -0.195607  0.115898
gossip     0.679132  0.057921
elder      0.233015  0.075071
num_seeds  0.068650  0.025823
gossip_ct  0.043503  0.005607
elder_ct  -0.007236  0.008028

Column 4: First Stage (Elder)
              Coef.   Std.Err.
Intercept  0.080916  0.110997
gossip     0.306066  0.076380
elder      0.834435  0.045693
num_seeds -0.034687  0.026057
gossip_ct  0.009250  0.006169
elder_ct   0.026628  0.009095
```

```
Column 5: IV
                              Parameter Estimates
==============================================================================
                   Parameter  Std. Err.     T-stat    P-value    Lower CI    Upper CI
------------------------------------------------------------------------------
const                 2.2745     1.0900     2.0867     0.0369      0.1381      4.4109
num_seeds            -0.4615     0.2647    -1.7433     0.0813     -0.9804      0.0574
gossip_ct            -0.0994     0.0562    -1.7686     0.0770     -0.2095      0.0108
elder_ct              0.2865     0.1035     2.7678     0.0056      0.0836      0.4894
atleast_gossip_1      1.8775     0.9829     1.9101     0.0561     -0.0490      3.8040
atleast_elder_1      -0.9481     0.5787    -1.6383     0.1014     -2.0823      0.1861
==============================================================================
```

Figure 2

Quantile treatment effects successfully replicated



Note: The intercept α(u) (not shown) corresponds to the random treatment group.

Study 2: Haryana Immunization RCT

521 villages · 4 arms · Outcome = children vaccinated per monthly camp

Context: Haryana — 7 low-immunization districts · Full immunization ≈ 40% at baseline · 1-year intervention (Feb 2017–Mar 2018)

T1
Random
6 random household heads

T2
Gossip
"Who spreads info widely?" (17 surveyed)

T3
Trusted
"Who do villagers trust?" (17 surveyed)

T4
Trusted Gossip
Both criteria combined

TABLE 3
Haryana immunization programme, communication treatment effect

	(1) Children received Penta1	(2) Children received Penta2	(3) Children received Penta3	(4) Children received measles	(5) Children attended session
Gossip	1.017	1.022	1.030	1.078	4.903
	(0.603)	(0.561)	(0.523)	(0.500)	(2.503)
Trusted	0.261	0.302	0.490	0.439	1.849
	(0.486)	(0.448)	(0.418)	(0.408)	(2.047)
Trusted gossip	0.479	0.526	0.514	0.444	2.376
	(0.470)	(0.429)	(0.396)	(0.376)	(1.917)
Observations	6697	6697	6697	6697	6712
Villages	521	521	521	521	521
Mean (Random seeds)	4.31	4.06	3.71	3.53	18.11
Gossip = Random (pval.)	0.092	0.069	0.049	0.032	0.051
Gossip = Trusted (pval.)	0.176	0.168	0.268	0.182	0.192
Gossip = Trusted Gossip (pval.)	0.343	0.338	0.281	0.166	0.271

	(1) Children received Penta1	(2) Children received Penta2	(3) Children received Penta3	(4) Children received Measles	(5) Children attended session
Gossip	1.056	1.056	1.060	1.099	5.052
	(0.604)	(0.563)	(0.525)	(0.501)	(2.509)
Trusted	0.250	0.295	0.486	0.436	1.821
	(0.486)	(0.449)	(0.419)	(0.409)	(2.052)
Trusted gossip	0.474	0.535	0.528	0.452	2.423
	(0.471)	(0.432)	(0.400)	(0.378)	(1.934)
SMS blast 33%	0.799	0.801	0.750	0.516	3.507
	(0.547)	(0.515)	(0.484)	(0.456)	(2.293)
SMS blast 66%	0.024	0.144	0.184	0.111	0.723
	(0.535)	(0.504)	(0.478)	(0.466)	(2.338)
Observations	6697	6697	6697	6697	6712
Villages	521	521	521	521	521
Mean (Random seeds)	4.31	4.06	3.71	3.53	18.11
Gossip = SMS Blast 33% (pval.)	0.746	0.725	0.656	0.382	0.643
Gossip = SMS Blast 66% (pval.)	0.214	0.236	0.226	0.153	0.211
Gossip = Random (pval.)	0.081	0.061	0.044	0.029	0.045
Gossip = Trusted (pval.)	0.153	0.148	0.241	0.168	0.169
Gossip = Trusted Gossip (pval.)	0.309	0.319	0.272	0.16	0.256

Table 2

Summary statistics of Haryana Immunization RCT

```
======================================================================
TABLE 2 - PANEL A: NOMINATION STATISTICS
======================================================================

Number of nominations (per village)
  Gossip                19.915 (8.585)
  Trusted               20.313 (8.670)
  Trusted Gossip        19.993 (11.351)

Nominations for top 6 individuals
  Gossip                11.217 (4.576)
  Trusted               10.560 (4.265)
  Trusted Gossip        10.769 (5.575)

======================================================================
TABLE 2 - PANEL B: SEED CHARACTERISTICS
======================================================================

Variable                                            Random          Gossip          Trusted        T.Gossip
------------------------------------------------------------------------------------------------------
Refused to participate                      0.186 (0.389)   0.165 (0.372)   0.219 (0.414)   0.175 (0.380)
Age                                        49.233 (14.617)  48.569 (14.347)  52.040 (14.130)  48.890 (14.082)
Female                                      0.054 (0.227)   0.108 (0.311)   0.055 (0.228)   0.098 (0.297)
Education (years)                           6.980 (4.280)   8.499 (3.966)   8.116 (4.073)   8.753 (3.930)
Owns land                                   0.586 (0.493)   0.675 (0.469)   0.680 (0.467)   0.687 (0.464)
Wealth index                                0.183 (0.098)   0.218 (0.121)   0.217 (0.114)   0.226 (0.120)
Hindu                                       0.705 (0.456)   0.731 (0.444)   0.684 (0.465)   0.736 (0.441)
Muslim                                      0.084 (0.278)   0.090 (0.286)   0.080 (0.272)   0.071 (0.257)
SC/ST                                       0.188 (0.391)   0.167 (0.373)   0.135 (0.342)   0.165 (0.371)
OBC                                         0.193 (0.395)   0.211 (0.409)   0.192 (0.394)   0.172 (0.378)
Panchayat member                            0.106 (0.308)   0.320 (0.467)   0.259 (0.438)   0.300 (0.459)
Numberdaar/Chaukidaar                       0.091 (0.288)   0.295 (0.456)   0.204 (0.403)   0.269 (0.444)
Interacts very often                        0.263 (0.441)   0.455 (0.498)   0.371 (0.483)   0.444 (0.497)
Participates very often                     0.264 (0.441)   0.457 (0.499)   0.371 (0.483)   0.445 (0.497)
Aware of immunization camps                 0.687 (0.464)   0.758 (0.428)   0.689 (0.463)   0.762 (0.426)
Aware of ANMs                               0.432 (0.496)   0.646 (0.479)   0.574 (0.495)   0.622 (0.485)
Aware of ASHAs                              0.605 (0.489)   0.794 (0.404)   0.706 (0.456)   0.780 (0.415)

Observations per treatment arm:
commtreat_arm
Random            570
Gossip            648
Trusted           712
Trusted Gossip    674
dtype: int64
```

Regression Results: Table 3

Estimates of communication treatment effects

TABLE 3 – PANEL A – Without SMS

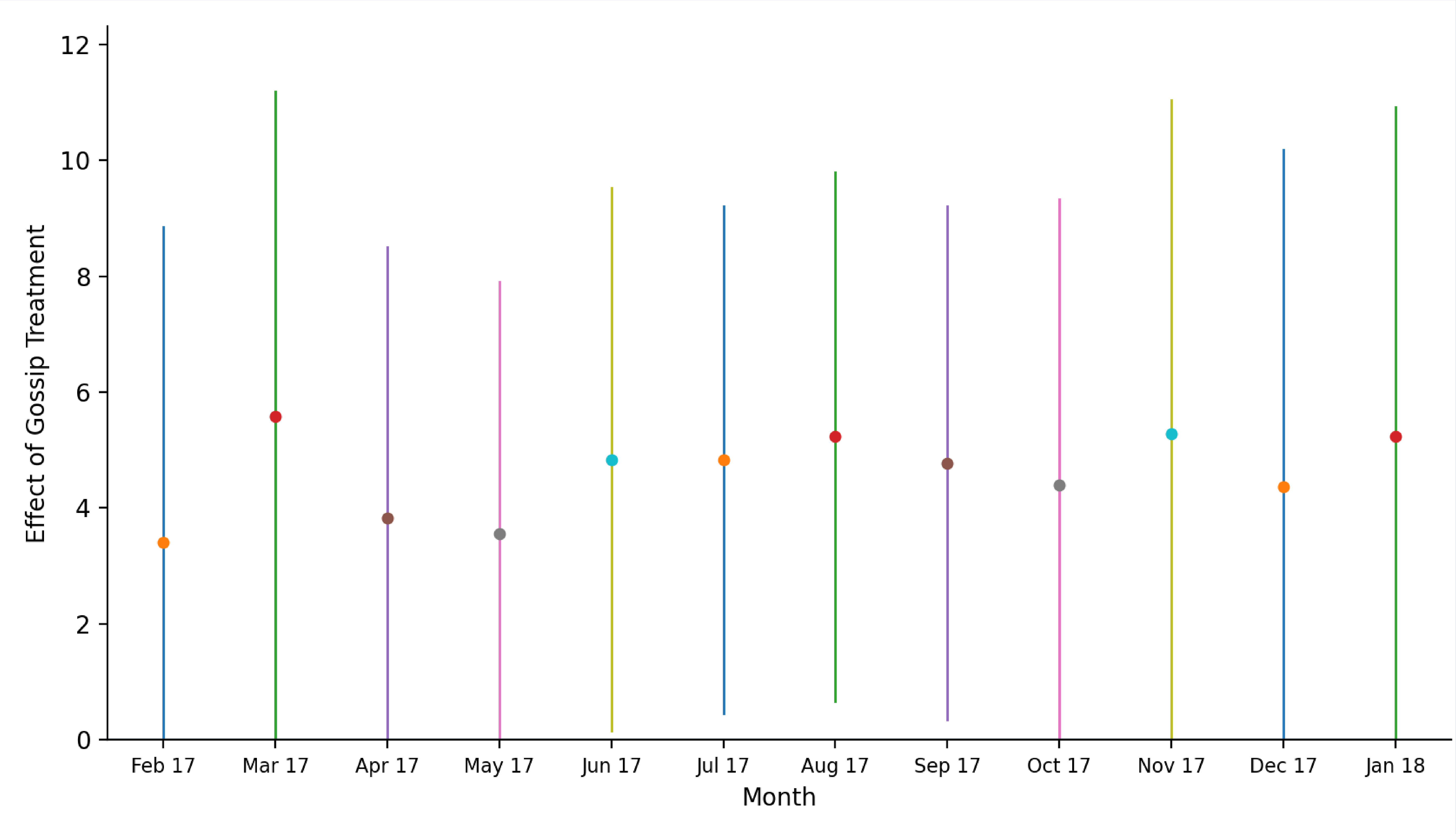
	Penta1	Penta2	Penta3	Measles	Any shot
Gossip	1.074	0.937	0.948	1.040	4.518
	(0.581)	(0.545)	(0.506)	(0.491)	(2.415)
Trusted	0.393	0.394	0.545	0.512	2.086
	(0.500)	(0.462)	(0.432)	(0.418)	(2.080)
Trusted Gossip	0.637	0.615	0.638	0.593	2.984
	(0.493)	(0.448)	(0.416)	(0.400)	(2.009)
Observations	6690	6690	6690	6690	6705
Villages	521	521	521	521	521
Mean (Random seeds)	4.31	4.06	3.72	3.53	18.12
Gossip = Random (pval.)	0.065	0.086	0.061	0.034	0.061
Gossip = Trusted (pval.)	0.244	0.322	0.434	0.286	0.319
Gossip = Trusted Gossip (pval.)	0.450	0.549	0.535	0.348	0.497

TABLE 3 – PANEL B – With SMS Controls

	Penta1	Penta2	Penta3	Measles	Any shot
Gossip	1.077	0.940	0.952	1.042	4.535
	(0.581)	(0.544)	(0.505)	(0.491)	(2.414)
Trusted	0.407	0.412	0.561	0.527	2.152
	(0.508)	(0.470)	(0.439)	(0.424)	(2.113)
Trusted Gossip	0.642	0.625	0.646	0.605	2.940
	(0.497)	(0.452)	(0.420)	(0.404)	(2.027)
SMS blast 33%	0.410	0.442	0.438	0.281	1.685
	(0.492)	(0.454)	(0.423)	(0.399)	(2.014)
SMS blast 66%	0.145	0.202	0.171	0.186	0.727
	(0.462)	(0.425)	(0.401)	(0.391)	(1.932)
Observations	6690	6690	6690	6690	6705
Villages	521	521	521	521	521
Mean (Random seeds)	4.31	4.06	3.72	3.53	18.12
Gossip = SMS 33% (pval.)	0.329	0.426	0.384	0.184	0.316
Gossip = SMS 66% (pval.)	0.155	0.223	0.170	0.116	0.161
Gossip = Random (pval.)	0.064	0.084	0.060	0.034	0.060
Gossip = Trusted (pval.)	0.250	0.333	0.446	0.295	0.327
Gossip = Trusted Gossip (pval.)	0.448	0.553	0.536	0.352	0.497

Replication: Figure 3

Effect of the “gossip tretament” on the number of children who attended Haryana immunization camps



ML Extension:

Predicting Optimal Seeds

Without Network Data

Extension: Predicting Optimal Seeds Without Network Data

ML models trained on cheap observables to predict diffusion centrality

Motivation: Can we skip both network surveys AND gossip surveys — and use cheap demographics to predict who is central?

Setup

Data: 6,466 households, 33 Karnataka villages
Target: DC_std (village-standardised diffusion centrality)
Features: GPS, leader, electricity, land, caste, rooms, gender, business…
Validation: 5-fold GroupKFold (whole villages held out)

Why is R^2 so low?

This is mathematically expected — not a flaw. Diffusion centrality depends on the entire network: who you're connected to, indirect paths, and clustering patterns. These are fundamentally unobservable from household characteristics. Observable traits can only proxy for these connections imperfectly, so most variation in DC remains unexplained.

CV Results (village-held-out GroupKFold)

Model	CV R^2	Spearman ρ
Lasso (CV α)	~0.065	~0.308
Ridge (CV α)	~0.065	~0.308
Random Forest	~0.031	~0.266
Gradient Boosting	~0.036	~0.269

Why does Spearman ρ (~0.27) matter more?

The policy problem is not to predict exact DC values — it is to identify who is more central. Spearman measures ranking accuracy. With ρ ≈ 0.27, the model correctly orders two random households ~62% of the time. For top-k selection, even a weak positive signal translates into large gains — this is the selection amplification effect.

Interpreting the Results

Why is adjusted R^2 ~0.03 not a failure?

R^2 measures level prediction accuracy. Diffusion centrality depends on unobserved network links — indirect paths, clustering, topology. Observable traits capture only the determinants of network formation (wealth, caste, geography), not the network itself. Low adjusted R^2 is mathematically inevitable, not a model flaw.

Why do linear models perform better?

With only 33 villages and a weak signal, tree models overfit village-specific noise. GroupKFold forces generalization to new villages — a regime where linear models' bias–variance trade-off is more favorable. Structure beats flexibility here.

Why does ρ ~0.24 matter for policy?

The decision problem is top-k selection, not point estimation. Spearman ρ directly measures ranking accuracy. With $\rho = 0.24$, the model correctly orders ~62% of household pairs — enough to systematically shift the selected seeds toward higher-centrality individuals.

How does this relate to the paper?

The paper shows gossip nominations approximate DC without full network data. This extension goes further: even without gossip surveys, ML on cheap observables achieves meaningful improvement. It extends the paper from 'cheap proxies' to 'purely observable-data proxies'.

Conclusions & Takeaways

RCT Evidence

Gossip nominees are significantly better seeds than random or elder-selected individuals — +22% immunization (Haryana), +40% calls (Karnataka). Effect is stable over 13 months.

Theory

Theorem 1 & 2: network gossip converges to diffusion centrality. Agents need only count — no knowledge of network topology required. Boundedly rational agents suffice.

Mechanism

Nominees are genuinely diffusion-central (LASSO picks DC over degree/eigenvector/leader). Network gossip percentile predicts nomination even with full individual fixed effects.

ML Extension

Cheap observable features (no network survey) can rank households by DC within villages (positive Spearman ρ). ML seeding beats random — useful when even gossip nomination is costly.

Appendix

Tables 6–9 are attached in the appendix and present detailed empirical results.

These tables report the extended specifications and supporting evidence for the main analysis.

They show that the core findings on diffusion and seed effectiveness remain consistent.

Table 6

Factors predicting nominations

```
================================================================================
TABLE 6 — BIVARIATE POISSON REGRESSIONS
================================================================================

—— Panel A: Event Nomination ——
Variable                                  Coef (SE)
------------------------------------------------------
Diffusion Centrality                  0.608 (0.122)
Leader                                0.863 (0.299)
Geographic Centrality                -0.010 (0.231)
Observations                                   6466

—— Panel B: Loan Nomination ——
Variable                                  Coef (SE)
------------------------------------------------------
Diffusion Centrality                  0.611 (0.116)
Leader                                0.962 (0.278)
Geographic Centrality                -0.205 (0.177)
Observations                                   6466
```

Table 7

Factors predicting nominations

```
==========================================================================
TABLE 7 — MULTIVARIATE POISSON REGRESSIONS
==========================================================================

── Panel A: Event Nomination ──
Specification                       DC        Leader          Geo        N
--------------------------------------------------------------------------
Col1: DC only             0.608 (0.122)          ---          ---     6466
Col2: + Leader            0.556 (0.140) 0.621 (0.319)          ---     5733
Col3: + Geographic        0.609 (0.121)          --- -0.042 (0.262)     6466
Col4: Full model          0.556 (0.141) 0.624 (0.322) -0.042 (0.275)     5733

Post-LASSO (Panel A: Event Nomination)
Selected variables: ['DC_std', 'leader', 'gps_std']
DC_std          0.556 (0.141)
leader          0.624 (0.322)
gps_std         -0.042 (0.275)

── Panel B: Loan Nomination ──
Specification                       DC        Leader          Geo        N
--------------------------------------------------------------------------
Col1: DC only             0.611 (0.116)          ---          ---     6466
Col2: + Leader            0.552 (0.127) 0.720 (0.292)          ---     5733
Col3: + Geographic        0.614 (0.114)          --- -0.261 (0.202)     6466
Col4: Full model          0.551 (0.126) 0.741 (0.287) -0.303 (0.217)     5733

Post-LASSO (Panel B: Loan Nomination)
Selected variables: ['DC_std', 'leader', 'gps_std']
DC_std          0.551 (0.126)
leader          0.741 (0.287)
gps_std         -0.303 (0.217)
```

Table 8

Does network gossip differentially predict nominations?

TABLE 8

Does network gossip differentially predict nominations?

	Nominated	Nominated	Nominated	Nominated	Nominated	Nominated
	(1)	(2)	(3)	(4)	(5)	(6)
Percentile of network gossip j,i	0.256	0.259	0.332	0.336	0.051	0.054
	(0.090)	(0.090)	(0.048)	(0.048)	(0.022)	(0.021)
Observations	665,301	665,301	665,301	665,301	665,301	665,301
Dep. var mean	0.382	0.382	0.382	0.382	0.382	0.382
Respondent FE		✓		✓		✓
Rankee FE					✓	✓
Flexible controls for DC			✓	✓		

Table 9

Calls teceived by seed types

TABLE 9 – Calls received by seed type

	(1) Calls	(2) Calls	(3) Calls	(4) Calls/Seed	(5) Calls/Seed	(6) Calls/Seed
At least 1 gossip	6.645	5.574		1.637	1.370	
	(3.842)	(4.092)		(0.943)	(0.985)	
At least 1 elder	0.346	0.057		0.245	0.173	
	(3.578)	(3.553)		(0.920)	(0.906)	
At least 1 high DC seed		3.663	5.183		0.916	1.312
		(2.477)	(2.367)		(0.619)	(0.644)
Observations	66	66	66	66	66	66
Control group mean	5.469	5.469	8.515	1.410	1.410	2.073
At least 1 gossip = At least 1 elder (pval.)	0.255	0.333		0.312	0.400	
At least 1 gossip = At least 1 high DC seed (pval.)		0.728			0.723	
At least 1 elder = At least 1 high DC seed (pval.)		0.420			0.481	

ML Extension: Summary Table

Predicting optimal seeds without network data

```
======================================================================
FINAL SUMMARY — Extension: Predicting Seeds Without Network Data
======================================================================

[A] PREDICTION PERFORMANCE  (5-fold village-held-out GroupKFold)
  Model                            R² ±    sd   Spearman ρ ±    sd
  --------------------------------------------------------------
  Lasso (CV α)                  0.065 ± 0.027        0.308 ± 0.052
  Ridge (CV α)                  0.065 ± 0.026        0.309 ± 0.053
  Random Forest                 0.031 ± 0.041        0.266 ± 0.062
  Gradient Boosting             0.036 ± 0.030        0.269 ± 0.068

[B] TOP PREDICTIVE FEATURES (permutation importance, RF)
  ----------------------------------------
  num_rooms                 +0.0831  ████████████████
  scst                      +0.0608  ████████████
  leader                    +0.0252  █████
  land                      +0.0174  ███
  gps                       +0.0151  ███
  have_electricity          +0.0036
  private_electricity       +0.0015
  own_house                 +0.0008
  business                  +0.0003
  female                    +0.0003
  laborer                   +0.0001

[C] SEEDING SIMULATION  (k=6, leave-one-village-out, N villages)
  Strategy                      DC sum     Gain     t-p
  ------------------------------------------------------
  Random (baseline)             -0.334          —
  ML — cheap features            3.208  +3472.6%  0.000
  Geographic (GPS)              -0.837   +597.4%
  Leader status                  0.846  +1284.0%
  Oracle (true DC)              13.458  +12591.6%
```

THANK YOU!

Questions are appreciated